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A hybrid model for image forgery detection using deep learning with block and keypoint methods

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Digital images serve as crucial evidence in fields like forensics and medicine, yet their reliability is increasingly threatened by sophisticated forgery techniques such as copy-move. While traditional block-based, keypoint-based, or deep learning approaches exist, each faces challenges regarding geometric transformations, dataset dependency, or computational complexity. This paper presents a novel hybrid model, HDBK, integrating deep learning, block-based, and keypoint-based methods for forgery detection at both image and pixel levels. The proposed framework comprises three fundamental stages. First, a triple-architecture ensemble of deep learning networks identifies forgery images and generates localized heatmaps to highlight suspected regions. Next, an enhanced block-based method utilizing a Genetic Algorithm (GA) is employed. This stage uses the maximum number of matched keypoints between candidate blocks as a fitness function to identify regions similar to the suspected forgery blocks. Finally, keypoint descriptors (SIFT, SURF, and FAST) are applied to match features between the identified regions, achieving precise pixel-level localization. This hybrid approach effectively reduces the search space for optimization, enhancing accuracy while minimizing false positive rates. The HDBK model was rigorously evaluated on the CoMoFoD dataset, which includes forgeries under various geometric transformations and post-processing operations like blurring and JPEG compression. Experimental results demonstrate that the model outperforms state-of-the-art techniques, particularly in detecting challenging scenarios such as small-scale and smooth forgery regions. The synergy between deep feature maps and meta-heuristic optimization ensures a robust balance between computational efficiency and forensic integrity in real-world passive forensic applications.

Keywords Copy-move forgery, Block-based, Key point-based, Deep learning, Pixel level forgery detection, Image level forgery detection, Deep neural network

Images play a crucial role in information exchange. With the widespread availability of digital cameras and smartphones, capturing photos has become feasible at any time and in any location. An estimate shows that around 1.4 trillion digital photographs were taken in 2020 alone¹. With the development of image editing software, it has become easier to manipulate images. Image forgery refers to the manipulation of digital image content with minimal self-evident clues, often with the intention of hiding or inserting false information into the image. By forging images, the authenticity and integrity of the image's structure and texture are manipulated. If an image is manipulated as a document, it is no longer trustworthy^{2,3}.

Digital forgery images are detected using two main approaches: active and passive. In the active approach, pre-processing is applied to the images, then the information is embedded into the original images. Digital signature and watermarking are examples of the active approach^{4–6}. The active approach requires software, specific hardware, and the original image to add information to the image and to extract information from it. On the other hand, in the passive approach, also known as the blind approach, no image preprocessing is required. This approach detects whether an image is forgery or not by analyzing its content and structure, uncovering inconsistencies. The passive approach does not require any prior information. For this reason, it has attracted increased attention from researchers. Copy-move forgery, image splicing, image retouching, and object removal are forgery types detected using this approach^{7,8}.

Copy-move forgery is one of the most famous forgery methods that can be done using the passive approach. In this method, one or more parts of an image are copied and pasted into other regions of the same image. This is done to hide or add false information. Along with copying and pasting, geometric transformations (such as rotation and scaling), and post-processing operations (such as blurring, changing brightness, compression, and

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adding noise) might be performed. These transformations and operations cause the manipulated regions to be not easily recognizable by humans⁹. An example of a copy-move forgery image is shown in Fig. 1

Forgery detection algorithms can be divided into three main categories: block-based method, key points-based method, and deep learning method¹¹. In the block-based method, the image is divided into fixed-size overlapping or non-overlapping rectangular, circular, or arbitrary blocks. In contrast, the key points method does not involve dividing the image. In both methods, forgery detection occurs through four steps: image pre-processing, feature extraction, matching similar features, and post-processing. Each of these steps has different parameters, which need to be set through cross-validation. Feature extraction and feature matching are fundamental steps, while pre-processing and post-processing are optional steps and are not considered in some studies¹¹.

Forgery image detection using block-based methods suffer from high computational time and complexity. Additionally, they exhibit poor performance for forgery image detection with certain geometric transformations and post-processing operations. In contrast, the keypoint-based method demonstrates greater robustness to geometric transformations. However, it has poor performance to detect small and smooth forgery regions due to insufficient key points. Both block-based and keypoint-based methods involve some steps. Enhancing forgery detection accuracy in block-based and keypoint-based methods requires fine-tuning each step's parameters¹².

Deep learning method performs the forgery detection process during the training of various neural architectures automatically. Unlike the block-based and key point-based methods, which are only used to detect forgery regions, deep learning methods are used to detect forgery at both the image and pixel levels. The big challenge of deep learning methods is the need for a lot of data for training and the lack of suitable datasets in the field of forgery images. To solve these issues, the pre-trained networks on large datasets and transfer learning techniques are employed^{11–14}.

In this study, we innovatively propose a hybrid model that leverages deep learning, block-based, and key point-based methods called HDBK to detect forgery images and regions. Figure 2 shows a overall view of proposed model. In the proposed model, three pre-trained networks are initially concatenated. The output of the concatenation layer is fed to the global average pooling layer and the softmax layer, respectively, to detect healthy and forgery images. Next, the parameters of this network are fine-tuned with the desired dataset. Subsequently, this network, without training on ground truth images that describe the forgery region, utilizes only the heatmap image. This heatmap image is produced using the concatenated feature maps to identify suspected forgery images and regions. Following the detection of suspected forgery regions, a block-based method employing a metaheuristic algorithm and integrating three keypoint extraction algorithms is applied. These combined algorithms act as a fitness function to identify similar regions or blocks to the forgery regions. Then, the key points between the forgery and similar blocks are matched. By using this matching technique, a convex polygon is drawn to accurately identify the forgery region. Next, the convex polygon is refined through morphological operations. In summary, the contributions of the proposed model are as follows:

- (i) Proposing a hybrid model using deep learning, block-based and key point-based methods to detect forgery at both image and pixel levels.
- (ii) Using global average pooling instead of fully connected layers in deep networks to increase the dependency between feature maps and classes.
- (iii) Forgery detection under various geometric transformations and different post-processing operations.
- (iv) Integrating the block-based method with a genetic algorithm to accelerate the identification of similar blocks to the suspected forgery block.
- (v) Employing three robust keypoint extraction algorithms to obtain sufficient key points for forgery detection in small and smooth regions.
- (vi) Identification of forgery regions by using heatmap images created from the feature maps of three pre-trained networks, without relying on ground truth images specifying forgery regions.

The remainder of this paper is organized as follows: Section [Related work](#) discusses related works in block-based, key point-based, and deep learning methods. In Section [Basic concepts](#), the basic concepts, used networks and algorithms are presented. The proposed model is described in Section [Proposed model](#). Section [Experimental](#)



Fig. 1. An example of copy-move forgery image from CoMoFod dataset¹⁰.

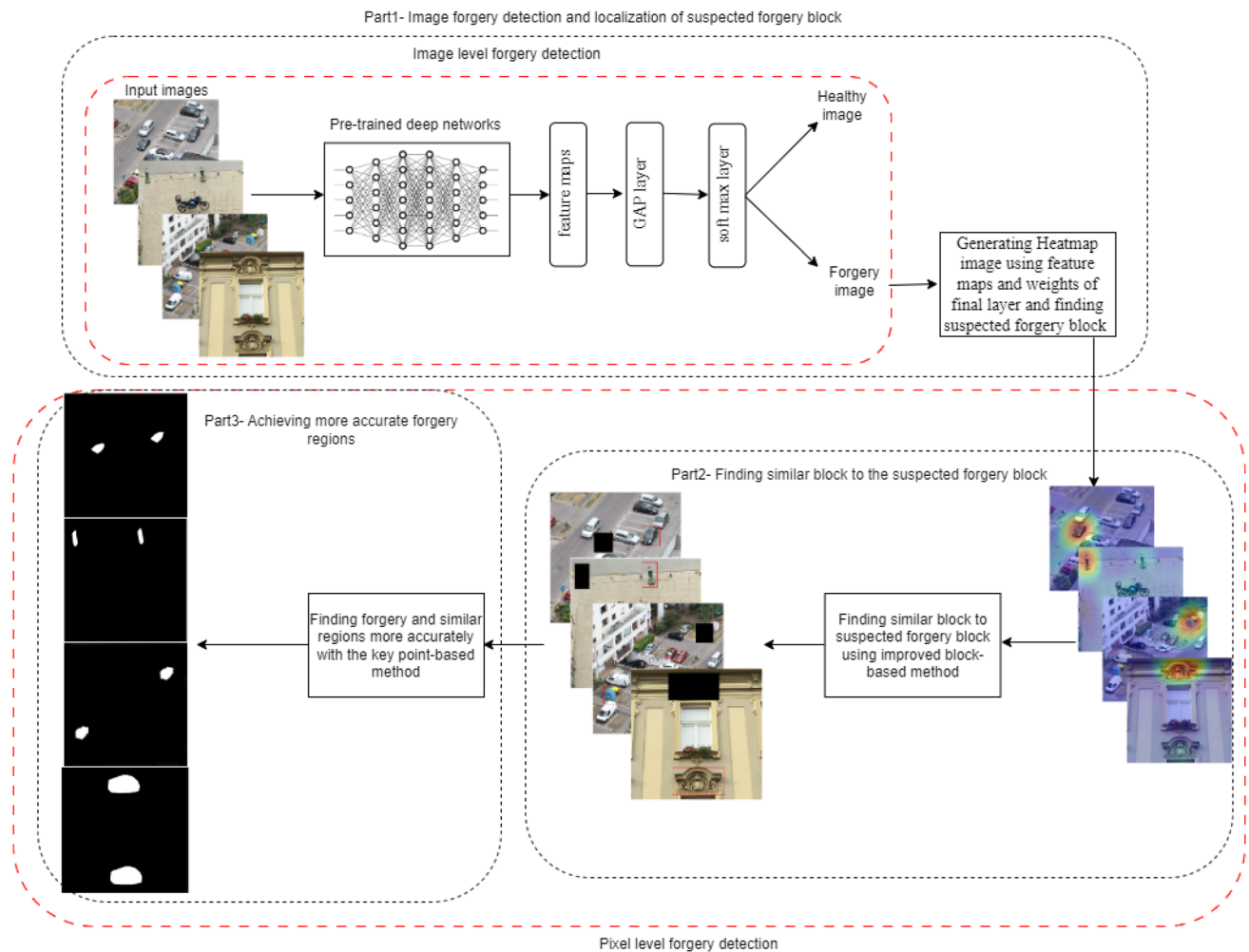


Fig. 2. Overall view of proposed HDBK model.

[results](#) summarizes the experimental results and compares them with other methods. Finally, we conclude our study in Section [Conclusion](#).

Related work

Copy-move forgery detection can be divided into three methods: deep learning, block-based and key point-based methods. This section reviews these three main methods.

Block-based forgery detection

There is a strong correlation between the copied and pasted regions in copy-move forgery images. These forgery regions can vary in shape and size. Searching the entire image space to identify these regions using the block-based method requires significant time and calculation. Forgery detection using a block-based method with Discrete Cosine Transformation (DCT) and Singular Value Decomposition (SVD) algorithms is investigated in¹⁵. In the pre-processing step, the image is resized to uniform size, and the RGB image is converted to a grayscale image. The grayscale image is then divided into overlapping square blocks of size 8x8 pixels. During the feature extraction step, the DCT algorithm is applied to each block, resulting in a feature vector. Next, the SVD algorithm is applied to these feature vectors to reduce their dimensions. Subsequently, the reduced feature vectors are categorized into forgery and healthy blocks using a support vector machine (SVM) classifier. Finally, the K-means clustering algorithm is used on forgery blocks to identify the forgery region.

In another block-based study¹⁶, in the pre-processing step similarly, the RGB image is converted to grayscale and the resulting grayscale image is divided into overlapped blocks. In the feature extraction step, a combination of DCT and eigenvalues is extracted from each block, yielding a feature vector. In the block matching step, the feature vectors from each block are sorted lexicographically, and similar blocks are identified using the Euclidean distance algorithm.

A block-based forgery detection method is investigated by using Babu et al¹⁷. During the pre-processing step, the RGB image is converted to grayscale. The resulting grayscale image then divided into overlapping blocks. In the feature extraction step, the polar complex exponential transform(PCET) algorithm is employed to extract

geometric invariant features from each block. Then, the Gradient Direction Pattern histogram is applied to reduce the dimension of extracted features. In the block matching step, the resulting feature vectors from each block are sorted, and similar blocks are identified using these histogram feature matrix rows. Finally, in the post-processing step, the windowing technique and morphological operators is used to filter out false matching blocks that are very similar but not tampered.

Keypoint-based forgery detection

In contrast to the block-based method, the keypoint-based method identifies feature points within high-entropy regions and describes local features without dividing the image. So, keypoint-based methods have greater speed compared to block-based methods.

A key-point based method for forgery detection is investigated in¹⁸. In the feature extraction step, keypoints are extracted from the image using the SIFT algorithm. In the feature matching step, the brute force matching technique is utilized to detect similar descriptor vectors between the copied and original regions in the forgery image. After identifying matching pairs, the density-based data clustering algorithm known as DBSCAN is applied to the keypoint spatial locations, grouping spatially close keypoints to detect forgery regions. Eventually, in the post-processing step, outlier detection techniques are employed to reduce a falsely matched pairs.

In another keypoint based method study¹⁹, the pre-processing step involves removing noise from images using Gaussian blurring and Wiener filtering. During the feature extraction step, keypoints are identified in textured areas using features from the Accelerated Segment Test (FAST) and Binary Robust Independent Elementary Features (BRIEF) algorithms. Additionally, keypoints for uniform and smooth regions are detected using the Scale Invariant Feature Transform (SIFT) algorithm. In the feature matching step, features are matched using a generalized second nearest neighbor approach. At the end, morphological processing followed by the Linear Spectral Clustering (LSC) algorithm is employed to enhance forgery localization.

A hybrid forgery detection method that utilizes the Discrete Cosine Transform (DCT) and Oriented FAST and Rotated BRIEF (ORB) algorithms is investigated in²⁰. In the pre-processing step, the RGB image is converted into a grayscale image, which is then divided into overlapping square blocks. During the feature extraction step, ORB and DCT algorithms are employed to extract key points and feature vectors from all the blocks, respectively. In the feature matching step, the extracted DCT features are matched based on Euclidean distance, while the extracted ORB keypoints are matched using the k-NN algorithm with Hamming distances to identify similar blocks.

Deep learning forgery detection

Recently, deep learning methods have become increasingly important in computer vision, particularly for detecting forgery images. These methods automatically extract meaningful features from data, avoiding the need for manual feature engineering. Two popular approaches for forgery detection based on deep learning networks are:

- End-to-end network or scratch network: In this approach, all layer from input to final layers are trained together using a large dataset.
- Pre-trained network: Here, pre-existing networks trained on large datasets serve as a starting point as known transfer learning techniques or feature extractor for new problems, especially when suitable datasets are scarce^{11,21}.

In the scratch approach, a network requires a lot of data and time for training all layers of networks. Unfortunately, in some research, there is no suitable dataset, making the use of this method almost impossible¹¹. For instance, deep learning-based forgery detection using the scratch approach is investigated in²². The smaller networks like VGGNet and MobileNet V2.0 are employed to detect copy-move forgery at the image level. During the training stage, two copy-move datasets using augmentation techniques are applied to train the parameters. Another example²³ involves using the VGG16 network to detect forgery images. For this purpose, a single forgery dataset and patches of images are used to train the network.

The pre-trained network is trained on a large dataset for a sufficient amount of time, and all its parameters are fine-tuned. These networks are capable of identifying important details from data and extracting meaningful features from new samples. There are two approaches for using pre-trained networks¹¹:

- Considering the extracted features as vectors and applying machine learning algorithms to classify them.
- Adding a new classifier on top of the pre-trained model to classify these features.

Using pre-trained networks as feature extraction for image forgery detection is investigated in²⁴. For this purpose, feature maps obtained from pre-trained models such as VGG16, MobileNet, and inceptionV3 are utilized as feature vectors. Next, these feature vectors are classified using different machine learning algorithms to detect copy-move forgery in image level.

The pre-trained VGG16 network that is trained on the extensive ImageNet dataset are used as part of a transfer learning approach in²⁵. Next, two new layers, the global average pooling (GAP) layer instead of default fully connected layer, and the softmax layer with two classes are added on top of this pre-trained network to categorize images into two forgery and healthy classes. Finally, parameters of some final layers of this network are fine-tuned using a forgery dataset.

Object detection networks and autoencoders can detect forgery. For example, the DenseNet-41 model is used as a backbone network with Mask R-CNN to detect copy-move forgery pixels in²⁶. The DenseNet model leverages features collected from very initial layers and propagates them to deeper layers. This robust feature

extraction enables accurate identification of forgery content. Next, the obtained features from DenseNet-41 are fed into the Mask-RCNN network for detecting and segmenting the forgery pixels.

In another deep learning method using autoencoder network²⁷, a CNN model is developed that utilizes multi-scale input with multiple stages of convolutional layers. These layers are divided into two blocks: the encoder and the decoder. In the encoder block, feature maps extracted from convolutional layers at various stages are combined and downsampled. In the decoder block, the extracted feature maps are combined and upsampled. Finally, a sigmoid activation function is used to classify pixels as either forgery or healthy.

Hybrid method for forgery detection

Based on the conducted studies, it can be concluded that the block-based method involves extensive computations and performs poorly against geometric transformations. Conversely, the keypoint method is resistant to geometric transformations but struggles to detect small forgery regions. Both of these forgery detection methods operate at the pixel level through multiple stages. In contrast, deep learning methods automatically detect forgery at both the image and pixel levels. However, a significant drawback of deep learning methods is their requirement for large datasets.

Some researchers use a combination of two methods to improve forgery detection. For example, the hybrid of the block-based method and keypoint based method using the SIFT algorithm is discussed in²⁸. The hybrid of the block-based method with a convolutional neural network for feature extraction of each block is used in²⁹. Forgery detection using the combination of convolutional neural network architecture with the CenSurE keypoint extraction algorithm is examined in³⁰. In this study, we employ a new hybrid model, integrating all three methods to mitigate their individual disadvantages and enhance forgery detection at both the image and pixel levels. Details of the proposed model are given in section [Proposed model](#).

Basic concepts

In this section, the used dataset, evaluation metrics, deep networks and metaheuristic algorithm are presented.

Dataset

Forgery detection, like other machine vision challenges, relies on standard datasets to evaluate. Some datasets like CMH³¹, SBU-CM16³², and CMFDDbgrip³³ include only forgery images, while others like MICC¹¹ and CASIA³⁴ include both original and forgery images. Additionally, some datasets like CoMoFod¹⁰ and COVERAGE³¹ provide binary images or ground truth images that identify regions of forgery. The focus of this research is copy-move forgery detection. Therefore, the descriptions of three famous datasets of copy-move forgery: MICC, CoMoFod and COVERAGE are explained below.

The MICC dataset is the oldest and most popular dataset in the field of copy-move forgery detection. It comprises four subsets: MICC-F220, MICC-F2000, MICC-F8multi, and MICC-F600. The forgery images in this dataset include geometric transformations such as rotation and scaling. However, there are no post-processing operations or ground truth images in the forgery images within this dataset. Additionally, the number of healthy and forgery images in each subset is limited or unbalanced^{11,31}.

CoMoFod as the well-known copy-move dataset consists of 10,000 images with 512×512 pixels. Of these, 5,000 images are forgeries, and 5,000 images are original. The copy-move method was used to create the forgery images. These forgery images are accompanied by attacks such as scaling, rotation, and various post-processing operations such as JPEG Compression (JC) with different quality factors, Image Blurring (IB) with $\mu = 0$ and $\sigma^2 = [0.009, 0.005, 0.0005]$, Noise Adding (NA) using the average filter with different sizes ($[3 \times 3, 5 \times 5, 7 \times 7]$), Brightness Change (BC), Color Reduction (CR), and Contrast Adjustments (CA) that are not present in other datasets¹⁰.

COVERAGE is third notable copy-move dataset. This dataset employs various geometric transformations for generating forgery images, such as scaling, rotation. However, it does not include post-processing operations for creating the forgery images. The COVERAGE dataset comprises a limited number of images: 100 original images and 100 forgery images, each with a resolution of 400×486 pixels. The size of the copy-move regions is relatively large, making the detection task less challenging. Each image is accompanied by two ground truth images: one for the duplicated region and one for the pasted region. Therefore, it is necessary to combine these two ground truth images before utilizing them^{11,31}. Table 1 provides a comparative summary of other existing benchmark datasets in the field of copy-move forgery detection, detailing their technical specifications and characteristics.

By comparing standard copy-move datasets, it is evident that the CoMoFod dataset provides a more comprehensive environment for evaluation. Unlike other benchmarks such as MICC or COVERAGE, which often lack precise binary ground-truth masks or diverse post-processing variations, CoMoFod includes a vast array of forgery images with different geometric transformations, multiple forgeries, and specifically, challenging smooth and small forged regions. The inclusion of extensive post-processing operations (e.g., JPEG compression, noise addition, and blurring) allows for a more rigorous and diverse stress test of the model's robustness at the pixel level. Consequently, due to its superior diversity and the availability of detailed ground truth for systematic benchmarking, this dataset was selected for evaluation in this study. Figure 3 shows three samples with their corresponding ground truth from the CoMoFod dataset.

Evaluation metrics

Forgery detection is investigated at two levels: image forgery detection and pixel forgery detection. These two levels are evaluated using standard evaluation metrics such as Accuracy, Recall, Precision, F1 score, False Positive Rate and Intersection over Union (IoU). These metrics are derived from the four components: True Positive, False Positive, False Negative, and True Negative. Table 2 defines these evaluation metrics^{31,35}.

Dataset	Format / Size	Images (A/T)	Shape	Geometric Changes	Post-processing	GT	Type
GRIP	PNG, 768×1024	A:80, T:3440	Arb.	S, R	NA, JC	Yes	CMF
CMFD	PNG, 458×344	A:50, T:970	Arb.	S, R	No	Yes	CMF
IMD	JPEG, 3000×2400	A:48, T:1344	Arb.	S, R	NA, JC	Yes	CMF
CVIP	BMP, 1000×700	A:70, T:970	Arb.	S, R	No	Yes	CMF
FAU	JPEG, 3000×2300	A:48, T:48	Arb.	All	NA, JC	Yes	CMF
CMH	PNG, 845×634+	T:216	Arb.	S, R	JC	Yes	CMF
SBU-CM16	JPEG, 800×580	T:240	Arb.	R	NA, JC, B	Yes	CMF
DVMM	BMP, 128×128	A:933, T:912	Square	-	No	No	CMF, S
CMFD-55	JPEG, 1000×700	A:50, T:50	Arb.	S, R	No	Yes	CMF
NIST16	JPEG, 500×500+	A:560, T:564	Arb.	S, R	JC	No	CMF, S, I
Columbia	TIFF, Various	A:183, T:180	Arb.	-	No	No	Splicing

Table 1. Summary of other common benchmark datasets in image forensics¹¹ (A: Authentic, T: Total Forged, Arb: Arbitrary, S: Scaling, R: Rotation, B: Blurring, NA: Noise Addition, JC: JPEG Compression, CMF: Copy-Move, S: Splicing, I: Inpainting, All: S, R, Translation, Distortion, and Combination).

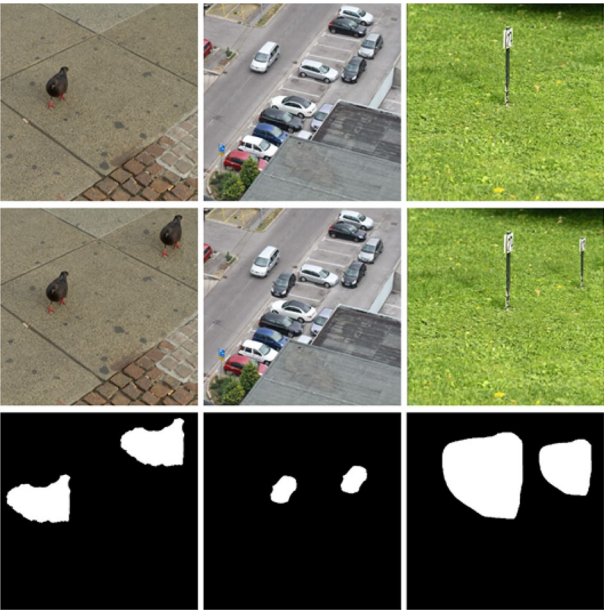


Fig. 3. Sample images of CoMoFod dataset: the first row: healthy images, the second row: copy-move forgery images, and the third row: ground truth images¹⁰.

In the image/pixel level forgery detection, TP represents the number of images/pixels that are correctly identified as forgery by the algorithm. FN expresses the number of images/pixels that are forgery, but the algorithm predicted them as healthy images/pixels. FP represents the number of healthy images/pixels that the algorithm falsely reported as forgery images/pixels, and TN represents the number of healthy images/pixels that were recognized as healthy by the algorithm.

Deep learning architectures

Since the advent of deep learning, numerous architectures have been developed and applied across various domains. Here, three well known architectures is briefly presented.

- VGG16 network: In 2014, the Visual Geometry Group at the University of Oxford proposed a convolutional neural network (CNN) architecture called the VGG16 model. In this model, there are 13 convolutional layers and 3 fully connected layers. One or more convolutional layers are followed by max-pooling layers, with progressively increasing depth. VGG16 is renowned for its simplicity and effectiveness, achieving strong performance on various computer vision tasks such as image classification and object recognition. This design enables the model to learn intricate hierarchical representations of visual features, leading to robust and accurate predictions³⁶.

Evaluation metrics	Formula
Accuracy	$ACC = \frac{TP+TN}{TP+FP+TN+FN}$
Recall	$R = \frac{TP}{TP+FN}$
Precision	$P = \frac{TP}{TP+FP}$
F1 score	$F1 = 2 * \frac{P * R}{P + R}$
False Positive Rate(FPR)	$FPR = \frac{FP}{FP+TN}$
Intersection over Union	$IoU = \frac{TP}{TP+FN+FP} = \frac{A \cap B}{A \cup B}$

Table 2. Different evaluation metrics.

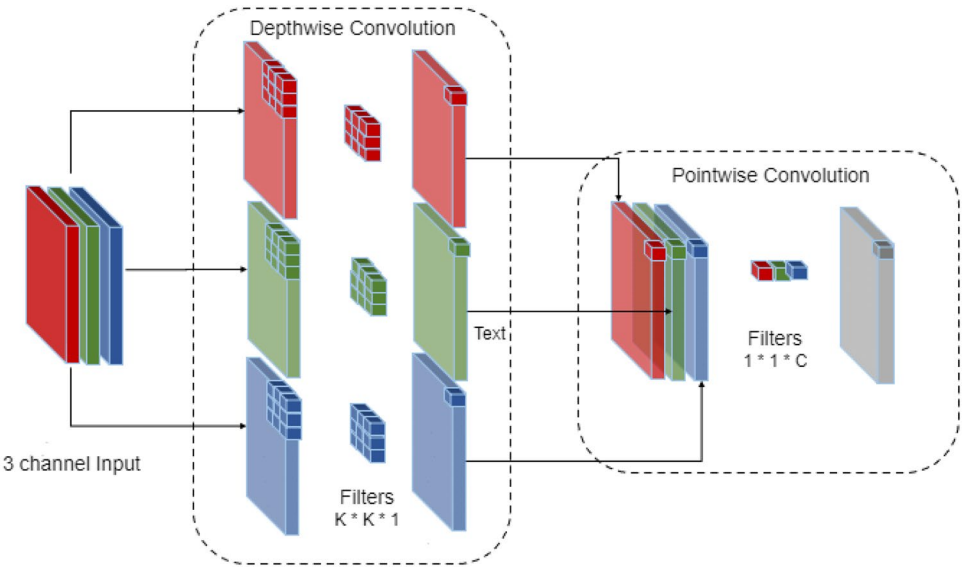


Fig. 4. The function of the depthwise separable convolution layer in MobileNet network³⁸.

- MobileNet network: MobileNet network is one of the networks developed by Google³⁷. This network is optimized for embedded systems such as mobile phones. The MobileNet network is a type of convolution-based network, but instead of the standard convolution layer, it uses a depthwise separable convolution layer at its core to build a lighter network. This layer uses two types of convolutions, Depthwise convolution and Pointwise convolution, to reduce calculation time. This type of convolution maintains accuracy at a desirable level by significantly reducing the number of parameters. Figure 4 shows the function of the depthwise separable convolution layer in MobileNet network.
- EfficientNetB0 network: The purpose of designed deep learning architectures is to make optimal use of available computing power and increase network performance. In architectures based on convolutional networks, there are three methods to increase accuracy: increasing the depth of the network, increasing the width of the network, and increasing the resolution of the input images. Enhancing any of these aspects will improve network performance. EfficientNet architecture is a convolutional neural network-based model that uses the concept of Compound model scaling and seeks to find the most efficient network according to the amount of computing power. The idea behind the EfficientNet architecture is to start with a baseline model and then improve it by increasing three dimensions: depth, width, and resolution. Setting these dimensions is an important and time-consuming process that requires a large search space^{39,40}. Figure 5 shows the function of the compound model scaling in EfficientNetB0 network.

Genetic algorithm

A Genetic Algorithm (GA)⁴¹ is a type of stochastic optimization method inspired by Darwin’s theory of natural selection. This population-based search algorithm is capable of solving complex optimization problems with unknown search spaces. The primary components of a GA include solution representation, selection, crossover, mutation operators, and fitness function evaluation. The genetic algorithm operates through the following steps:

- 1- Define the algorithm’s parameters, such as population size, maximum iterations, and fitness function, based on the specific problem.
- 2- Generate an initial population of candidate solutions randomly and set the iteration index to zero.

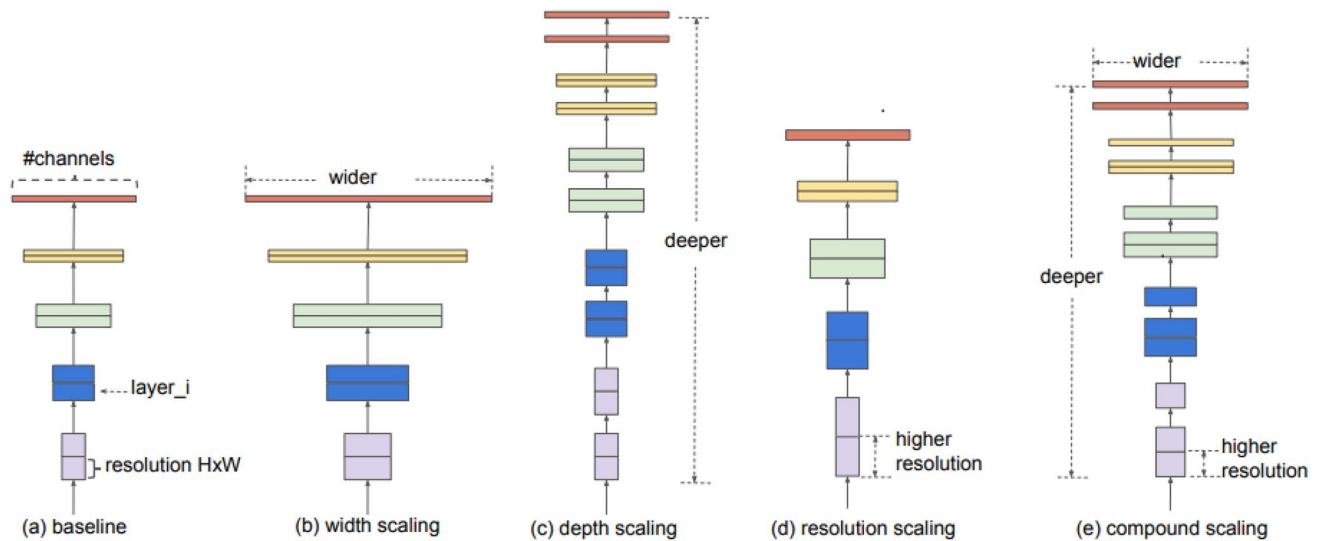


Fig. 5. The function of the compound model scaling in efficientNetB0 network⁴⁰.

- 3- Calculate the fitness value for each solution.
- 4- Select a subset of current solutions based on their fitness values. Apply crossover and mutation operators to these selected solutions to create new solutions.
- 5- Replace the old population with the new one and increment the iteration index.
- 6- If the iteration index reaches the maximum number of iterations, return the best solution from the population as the final solution; otherwise, repeat from step 3.

In forgery detection using the basic block-based method, all pairs of blocks, whether overlapping or non-overlapping and of various sizes, must be compared based on the features extracted from each block. This is an NP-hard problem. Therefore, in this study, the genetic algorithm is used to solve this issue. To identify forgery and similar blocks using the genetic algorithm, only a limited number of block pairs are compared based on the number of matched keypoints.

Proposed model

Forgery detection can be performed at two levels, Image-level and Pixel-level. In image-level forgery detection, the entire image is classified as either healthy or forgery. In pixel-level forgery detection, each individual pixel within the image is assigned to one of these two classes¹¹. Since forgery detection at both levels is important, we introduced a new hybrid model called HDBK, which uses deep learning, block-based, and key point-based methods to detect forgery at both levels. The HDBK model consists of three main parts, as illustrated in Fig. 2. In the first part, images are initially classified into healthy and forgery categories using pre-trained networks (which is the image level forgery detection). Next, a heatmap image is generated for each forgery image, and the suspected forgery block is identified on the heatmap. In the second part, a block-based method with a genetic algorithm is used to search for similar blocks to the suspected forgery block. Finally, in the third part, accurate forgery regions are detected. In this part key-point method matches keypoints of similar and forgery blocks (which is the pixel level forgery detection). In the following sections, details of the forgery detection process by using HDBK model at both the image and pixel levels are provided.

The first part of HDBK model: forgery detection at the image level

The forgery detection at the image level are depicted in Fig. 6. In the proposed model, we utilize the concept of a functional API to construct a new network. This network takes three inputs with color images and different branches, which combine three pre-trained networks. Among various available deep learning architectures, VGG16, MobileNet, and EfficientNetB0 demonstrated the best performance in detecting forgery images. VGG16 has been selected for its simple architecture, strong feature representation, and good performance in most computer vision tasks. MobileNet has been chosen for its low memory consumption, and EfficientNetB0 has been selected for its high accuracy and efficient computational power usage. The input images for all three branches have the same dimensions of $224 \times 224 \times 3$ and are fed into different layers of the three mentioned networks. The concept of transfer learning is employed to utilize the pre-trained weights of each network. Additionally, by fine-tuning the weights, only the last four layers of each network are involved in the training process using the desired dataset. At the end of the proposed network, the feature maps obtained from the final layers of the three networks are merged using the concatenate layer, resulting in a feature map with dimensions of $7 \times 7 \times 2816$. Following, the global average pooling layer is innovatively used instead of the fully connected layer. The GAP layer is chosen because it is a native layer that creates a strong dependency between the feature maps and each class. This layer is parameterless, which helps avoid the possibility of overfitting during the training process. Additionally, the GAP layer summarizes spatial information and is resistant to spatial transitions⁴².

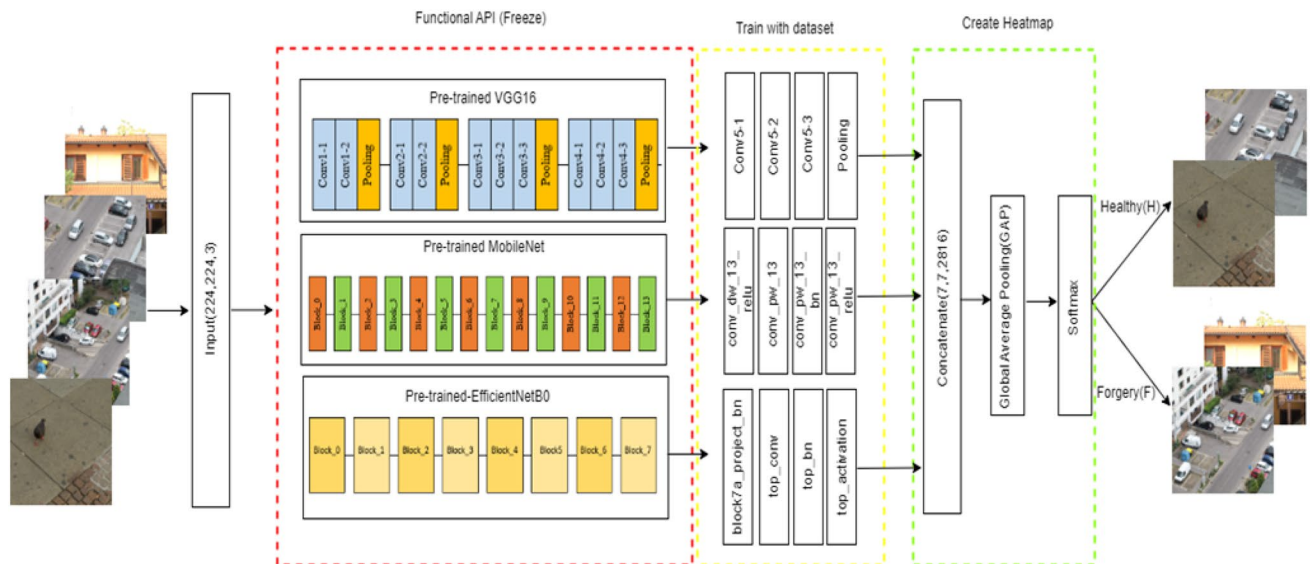


Fig. 6. The process of the first part of HDBK model: forgery detection at the image level.

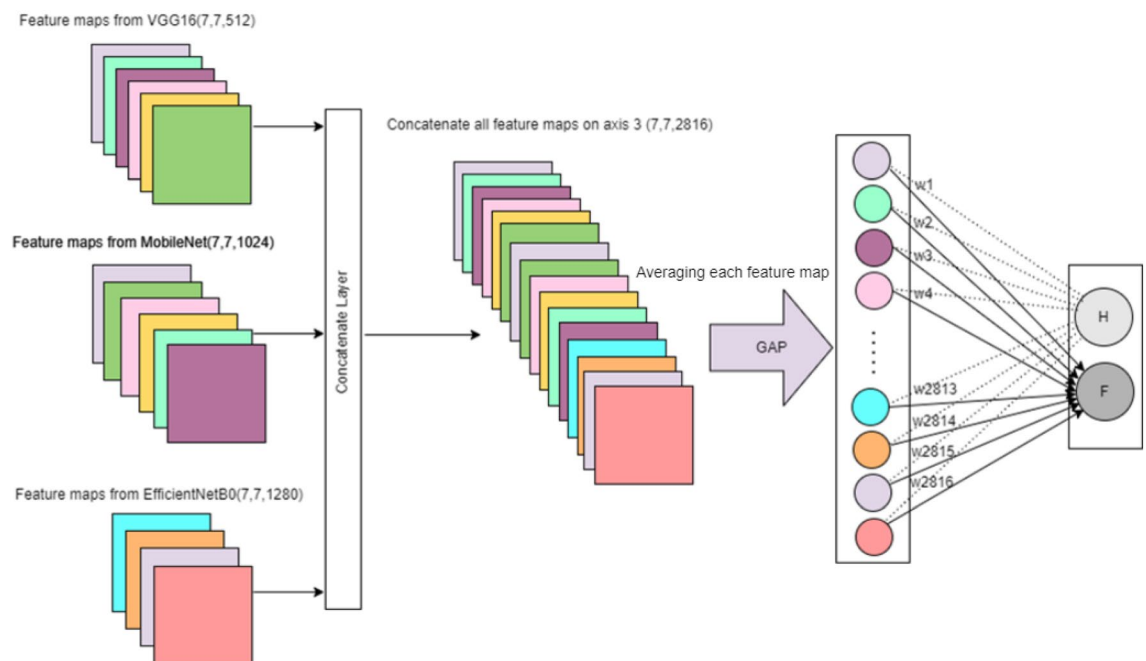


Fig. 7. Function of concatenate and GAP layers in the proposed network.

The function of the GAP layer is illustrated in Fig. 7. The input to this layer consists of the feature maps combined in the concatenate layer. The GAP layer transforms each of the merged feature maps into a single value using an averaging operation. Consequently, the output of the GAP layer in the proposed network is 2816 values, which are used as input in the discriminating layer. Finally, a soft max layer with two classes is placed at the end of the proposed network to classify the images into two classes: healthy and forgery.

Detecting the suspected forgery block

As shown in Fig. 7, each node in the GAP layer is connected to all the nodes in the softmax layer using unique weights(W_i). These weights are used to create a heatmap image for forgery image. A heatmap is a powerful tool in computer vision for solving classification problems⁴³. In this study, it has been used to detect forgery regions. This tool is useful for visualizing and identifying which parts of the image are important for decision-making.

The heat map image is created from the product of the combined feature maps in the concatenate layer and the weights calculated from the final layer. Since the goal is to identify forgery regions in a forgery image, the

weights between the nodes of the GAP layer and the forgery class node are used to construct the heat map. Next, by applying the maximum filter on the heat map images, the local peak points of the images are identified. Then, by applying thresholding on the neighbors of these maximum points and merging them, the block containing the forgery regions is determined. The process of generating a heat map image is shown in Fig. 8.

Forgery detection at the pixel level

Once the forgery image and suspected forgery block are identified at the first part of HDBK using the hybrid network, the pixel-level forgery detection process is conducted in two steps: 1- Finding a similar block to the suspected forgery block. 2- Achieving more accurate identification of forgery regions.

The second part of HDBK model: finding a similar block to the suspected block

In forgery detection using the basic block-based method, all pairs of image blocks must be compared based on their features to find similar block pairs. Consequently, this method can be time-consuming. In this study, to reduce the comparison time, an improved block-based method with genetic algorithms is presented, and only a limited number of blocks are investigated. Therefore, after identifying the suspected forgery block using the heat map image, a combination of block-based and key point-based methods and genetic algorithm is used to identify the similar block. Next, the detected forgery block is removed from the forgery image, and a similar block to the deleted block is searched using a genetic algorithm. Figure 9 illustrates an overview of the combined block-based method and genetic algorithm used to detect similar and forged blocks.

Firstly in genetic algorithm, an initial population of chromosomes is created. Each chromosome contained four genes: the first two genes represented the x and y coordinates of the upper left point of the block, and the last two genes represent the height and width of the block. Figure 10 shows the structure of chromosome. The fitness function of genetic algorithm is the inverse of the number of matched key points that is obtained from three keypoints extraction algorithms and the similar block is searched. Among the keypoint extraction algorithms, the SURE, FAST, and BRISK algorithms have been selected due to their high speed and accuracy in keypoint extraction, and resistance to geometric transformations, changes in illumination, contrast variations, and noise. Additionally, a combination of these three algorithms has been used to extract sufficient and appropriate keypoints in small and smooth forgery regions. The formula of the fitness function is given in Eq 1.

$$\text{Fitness Function} = \frac{1}{\text{The matched keypoints of blocks}} \quad (1)$$

The new chromosomes are created using the single-point crossover operation between the first and second genes and the uniform mutation operation by randomly generating the x or y coordinates of the selected chromosome. The specific parameter settings used for the genetic algorithm in this study, including population size, number of generations, and operator rates, are summarized in Table 3.

Finally, the best chromosome from the last iteration is considered as the similar block with the forgery block. Figure 11 shows the process of single point crossover and uniform mutation operations. The red blocks represent the results of the crossover operator, while the green block represents the result of the mutation operator.

It should be noted that some forgery images contain multiple forgeries. There are two methods to find similar blocks: 1) The genes of the chromosome should be modified so that, instead of discovering the x and y coordinates of one similar block, it searches for the x and y coordinates of several similar blocks. 2) Repeat the steps of the genetic algorithm multiple times until all similar blocks are found. Figure 12 shows the similar blocks searched by the hybrid of improved block-based and keypoint based methods for some forgery images.

The third part of HDBK model: achieving more accurate identification of forgery regions

After discovering the similar block to the forgery block, key points between the two blocks are matched using three key point extraction algorithms to identify similar objects. Therefore, after discovering the corresponding key points between pairs of blocks, a convex polygon is drawn using the discovered points of each block. The generated convex polygons are improved using morphological operations, and a binary mask is produced for each pair of forgery blocks and similar blocks. At the end, the created pair of binary masks is placed in the appropriate location on the forgery image and merged together. For the forgery image, a binary mask representing the location of the forgery and the similar regions is obtained. The process of identifying accurate forgery objects is summarized in Fig. 13.

To highlight the innovations and enhance the clarity of the proposed methodology, the step-by-step procedure of the HDBK model is presented in Algorithm 1.

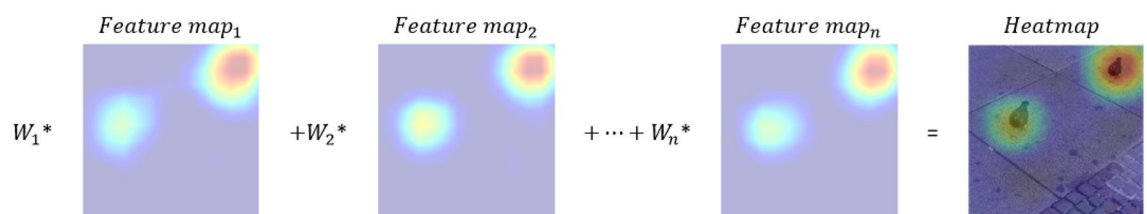


Fig. 8. The process of generating the heat map image.

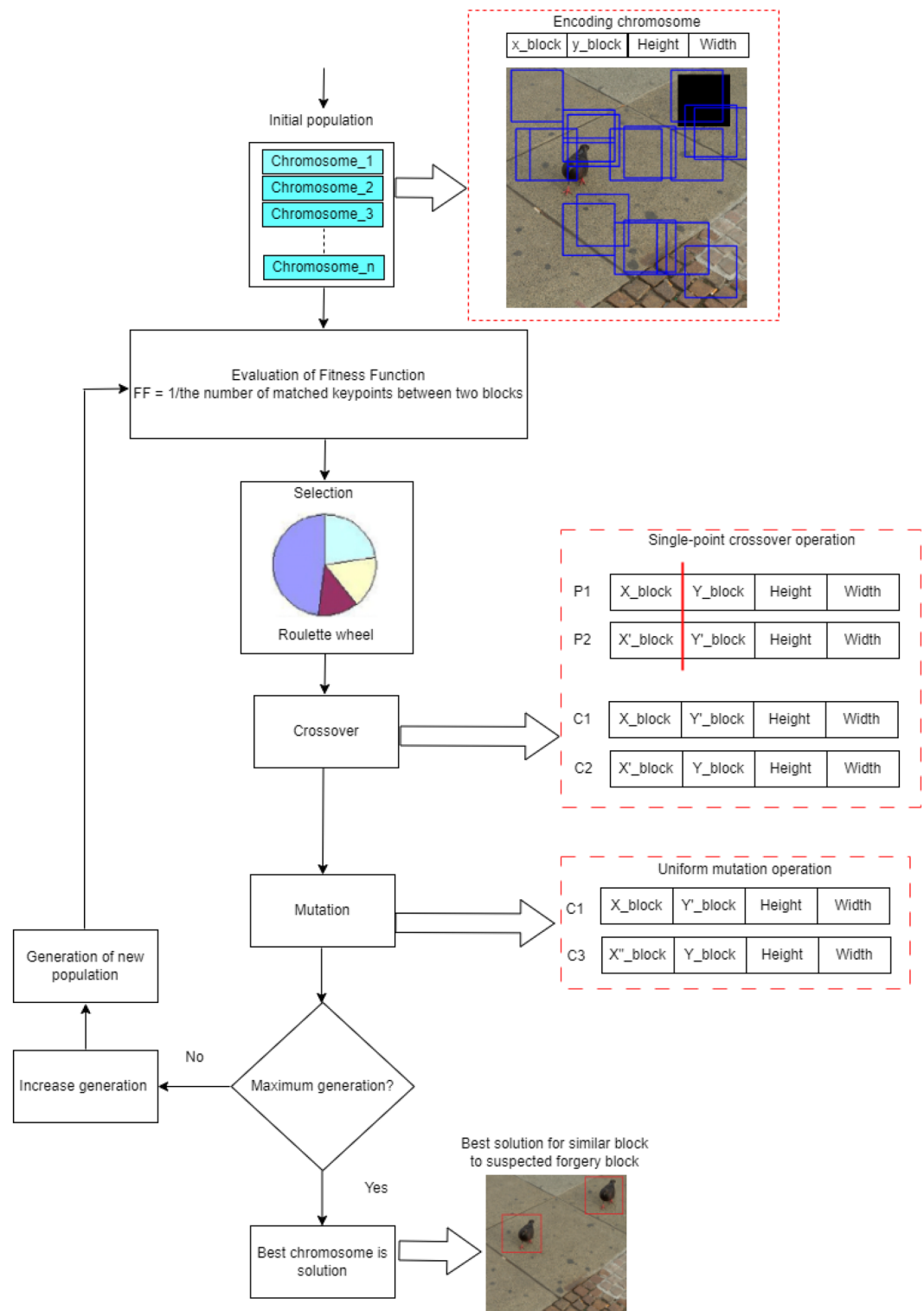


Fig. 9. Flowchart of used genetic algorithm to detect similar and forgery blocks.



Fig. 10. The structure of chromosome.

Population Size	Number of generations	Crossover rate	Mutation rate	Selection strategy
50	50	0.80	0.05	Roulette Wheel

Table 3. Parameters of genetic algorithm.

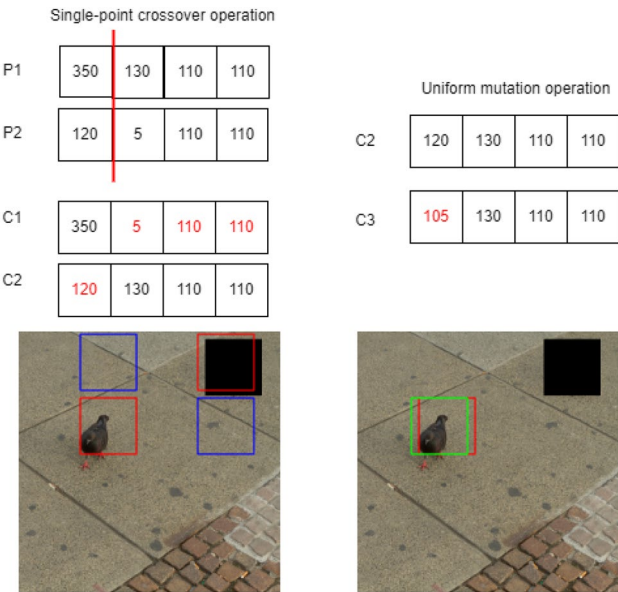


Fig. 11. The process of single point crossover and uniform mutation operations.

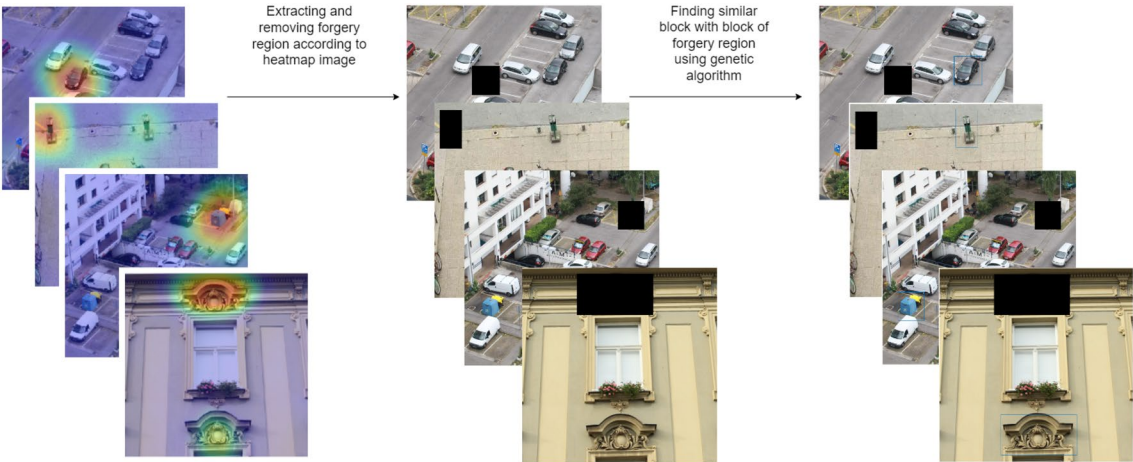


Fig. 12. The similar blocks searched by the improved block-based method.

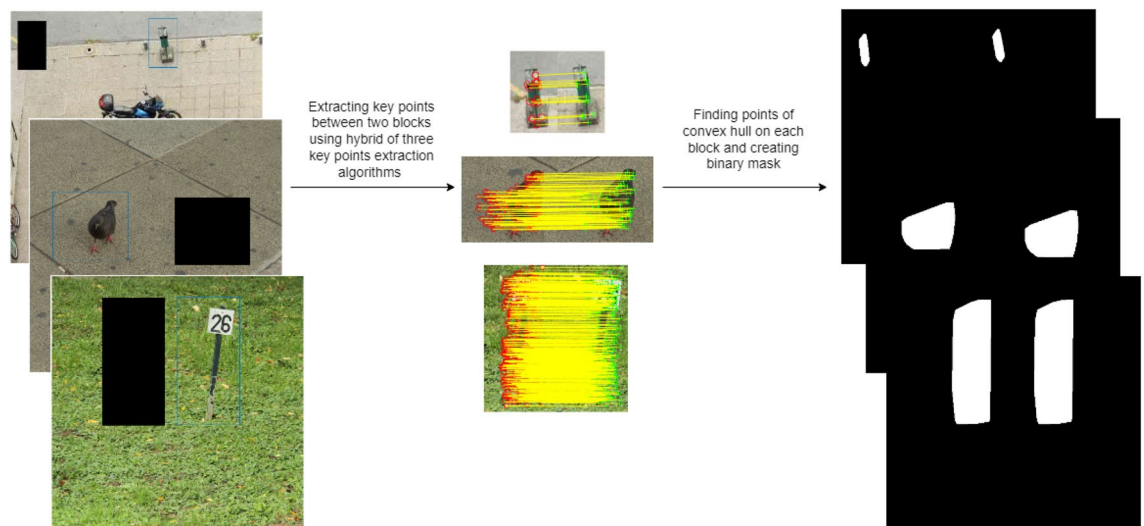


Fig. 13. Achieving more accurate identification of forgery regions.

-
- 1: **Input:** RGB Image I , GA parameters (P, G, C, M) , Weights W
 - 2: **Output:** Forgery Mask M_{final}
 - Part 1: Deep Learning & Heatmap Localization**
 - 3: $F_{concat} \leftarrow$ Concatenate feature maps of VGG16, MobileNet, EfficientNetB0
 - 4: $H \leftarrow$ Generate Heatmap using F_{concat} and Softmax top layer weights
 - 5: $H_{filtered} \leftarrow$ Apply Max Filtering on H to locate peak suspicion
 - 6: $B_{suspect} \leftarrow$ Extract block at coordinates of $\max(H_{filtered})$
 - Part 2: Genetic Algorithm for Block Matching**
 - 7: $Population \leftarrow$ Initialize P chromosomes as $[x_i, y_i]$
 - 8: **for** $gen = 1$ **to** G **do**
 - 9: **Fitness Evaluation:** (Max matching keypoints)
 - 10: $Fit(Ch_i) \leftarrow$ Count matches($B_{suspect}, Block(x_i, y_i)$)
 - 11: $Parents \leftarrow$ Roulette Wheel Selection based on Fit
 - 12: $Offspring \leftarrow$ Single-point Crossover on $[x, y]$
 - 13: **Mutation:** (Randomly change one or two genes)
 - 14: **if** $rand() < Mutation_Rate$ **then**
 - 15: $type \leftarrow$ Randomly select from $\{x, y, both\}$
 - 16: Update offspring with new random coordinates
 - 17: **end if**
 - 18: **end for**
 - 19: $B_{similar} \leftarrow$ Block with $\max(Fit)$ (Global best solution)
 - Part 3: RANSAC Verification & Localization**
 - 20: $K_{matches} \leftarrow$ Match points (SURF, FAST, BRISK) between $B_{suspect}, B_{similar}$
 - 21: $Inliers \leftarrow$ Apply **RANSAC** to remove false matches and outliers
 - 22: $Poly \leftarrow$ Draw Convex Polygon surrounding $Inliers$
 - 23: $M_{final} \leftarrow$ Refine $Poly$ using morphological opening/closing
 - 24: **return** M_{final}
-

Algorithm 1. Proposed Hybrid HDBK Model for Image Forgery Detection

Experimental results

The investigation of the proposed model at both image and pixel levels is presented in this section. It should be noted that all the code related to the proposed hybrid model is written using the python programming language and the open-source libraries tensorflow and keras. The implementation of the proposed model is performed on GPU in the colab environment.

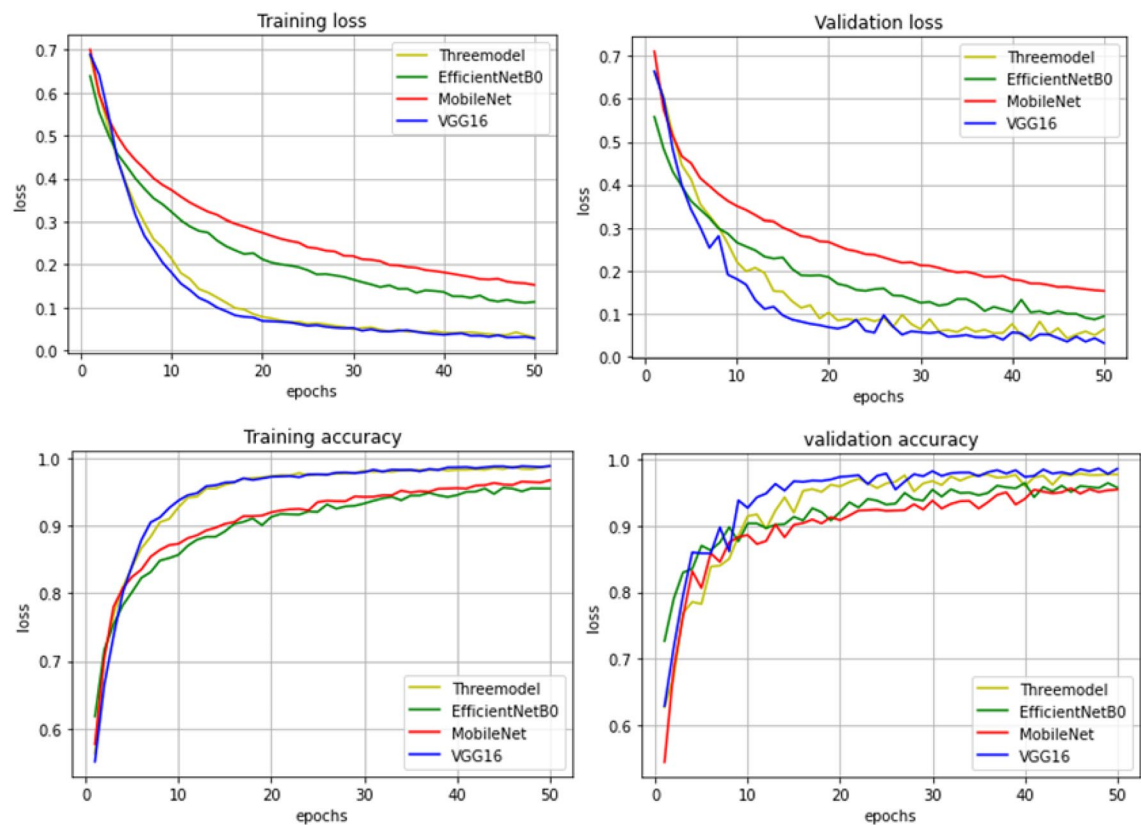


Fig. 14. Accuracy and loss function for image forgery detection on the CoMoFod dataset.

Parameters	Value
Epoch	50
Batch size	32
Learning rate	0.00001
Optimizer	Adam
Loss function	Categorical-cross entropy
Activation	Soft max

Table 4. Parameters of the proposed hybrid network.

Evaluation of forgery detection results at the image level

To evaluate forgery detection at the image level, the CoMoFod dataset is used, which contains both healthy and copy-move forgery images. A positive aspect of this dataset is the inclusion of various geometric transformations and post-processing operations. These operations have been performed on all images in the CoMoFod dataset, both healthy and forgery.

According to Fig. 6, this study introduces a hybrid network using the concept of API to detect forgery at the image level. A new three-branch network is designed utilizing three pre-trained networks: VGG16, MobileNetV2, and EfficientNetB0. To leverage the ensemble learning/feature fusion concept, a concatenate layer is added to merge the feature maps of these three networks. At the end of the network, a global average pooling layer and a softmax layer are used, respectively, to classify images into two groups: healthy and forged. After designing the hybrid network, the final five layers are trained using the CoMoFod dataset to fine-tune their parameters. It should be noted that the number five was chosen through cross-validation, balancing performance and training time. During the training process, 70% of the images are used as training data, 15% as validation data, and 15% as testing data. The training process of the proposed hybrid network and its constituent networks is conducted separately for 50 iterations using the Adam optimizer. Other parameters of the networks, along with the diagrams of the loss function and accuracy for all four mentioned networks, for both training and validation data, are presented in Fig. 14 and Table 4, respectively.

Furthermore, an ablation study was conducted to evaluate the individual contribution of each CNN backbone. As shown in Table 5, while the individual pre-trained models such as MobileNetV2, EfficientNetB0, and VGG16 achieve high performance, the proposed hybrid API model yields the highest accuracy (98.76%)

Model Architecture	Accuracy (%)	Loss Function
Pre-trained MobileNetV2 + GAP layer	95.45	0.1527
Pre-trained EfficientNetB0 + GAP layer	97.40	0.0486
Pre-trained VGG16 + GAP layer	98.50	0.0310
Proposed API Model (Hybrid)	98.76	0.0286

Table 5. Ablation study of individual backbones vs. the proposed hybrid API model on CoMoFoD dataset.

and the lowest loss (0.0286). This demonstrates that the fusion of feature maps from multiple architectures significantly enhances the robustness of image-level forgery detection.

Evaluation of forgery detection results at the pixel level

After training the networks, the images in the test set are classified into two classes: healthy and forgery. Next, the heat map for the forgery images is prepared using the hybrid network and its constituent networks. In the VGG16 network, the feature maps are obtained from the block5_pool layer. In the MobileNet network, the feature maps come from the conv_pw_13_relu layer. In the EfficientNetB0 network, the feature maps are from the top_activation layer. In the proposed network, a combination of the feature maps from the concatenated layer is obtained. Finally, these feature maps from each of the four mentioned networks, along with the weights between the nodes of the GAP layer and the forgery class node, are used to create a heat map in the forgery image. Figure 15 shows the heat map of some examples of CoMoFod dataset images using the mentioned four networks.

According to the diagrams in Fig. 14, it can be concluded that the VGG16 network and the hybrid network have similar performance in image forgery detection. However, based on the generated heat maps in Fig. 15, it is clear that the combined network using integrated feature maps performs better than the separate networks in detecting the correct and more accurate forgery regions. In the following, to detect the forgery region, the local peak point is found by applying the maximum filter on the heat map resulting from the combined network. Then, using this point and applying threshold on it, neighboring points are discovered, and the block indicating the region of forgery is determined.

The block of forgery region marked on the heat map is removed from the forgery image. Using an improved block-based method with a genetic algorithm, the block that has the highest similarity in terms of the number of matched key points with the deleted block is considered as similar block. The genetic algorithm process is executed over 50 iterations, utilizing 50 chromosomes, with a crossover rate of 0.80 and a mutation rate of 0.05. In the following, the forgery pixels are determined more precisely through the combination of three key point extraction algorithms namely SIFT, SURF, and FAST applied between two forgery block and the identified similar block. Finally, by drawing a convex polygon using the discovered key points, a binary mask is created for the forgery image. Figure 16 shows the generated binary masks using hybrid model for several CoMoFod dataset images. The results of the proposed model at the pixel level are provided in Table 6.

In the proposed framework, the generated heatmaps serve as an intermediate localization stage designed to identify the coarse region of interest and candidate peak points. While heatmaps provide a visual representation of the model's focus, their reliability is quantitatively validated by the final binary masks. As demonstrated in Table 6, the high IoU (87.17%) and F1-score (91.85%) are obtained by comparing these refined results derived from the heatmap guidance directly with the ground-truth masks. This confirms that the heatmaps accurately align with the actual forged regions, providing a robust foundation for the subsequent genetic optimization and pixel-level refinement.

It is worth noting that the proposed HDBK model was implemented using the TensorFlow and Keras frameworks in Python. All experiments and simulations were conducted on the Google Colab cloud platform, utilizing a Tesla T4 GPU (16GB GDDR6) and 12.7 GB of System RAM. The computational workflow is divided into two distinct phases: an offline training phase for the triple-architecture ensemble, which requires approximately 150 minutes, and an online inference phase. During testing, the model demonstrates high efficiency by completing the detection and localization process in 26.95 seconds (comprising 14.35s for deep learning-based heatmap generation and 12.6s for the Genetic Algorithm-optimized search). These specifications confirm that the HDBK model is not memory-intensive and is compatible with standard hardware environments, making it a practical tool for digital forensic applications.

To evaluate the robustness and practical reliability of the proposed HDBK model, the method was extensively tested against various post-processing operations, including blurring, noise addition, JPEG compression, and contrast adjustments. The quantitative results of these comprehensive evaluations are summarized in Tables 7, 8, 9, 10.

A detailed analysis of the results in Tables 7 to 10 clarifies the operational boundaries of the HDBK model. The slight decline in metrics compared to the overall accuracy in Table 6 (97.34%) is a logical consequence of focusing exclusively on high-stress scenarios. In Table 7, increasing the blurring intensity ($\sigma^2 = 0.005$) leads to a marginal drop in accuracy to 94.85%, as the smoothed edges reduce the distinctness of the forgery boundaries. Regarding noise (Table 8), the 3×3 filter proves more disruptive than the 5×5 version, as fine-grained distortions more effectively mask local intensity gradients. Furthermore, a critical performance boundary is identified in Table 9 under extreme JPEG compression ($QF = 20$), where the FPR increases to 8.92%. This is attributed to the dominance of blocking artifacts, which mimic forensic traces and complicate the Genetic Algorithm's search for similar blocks. Similarly, the reduction of contrast dynamic range in Table 10

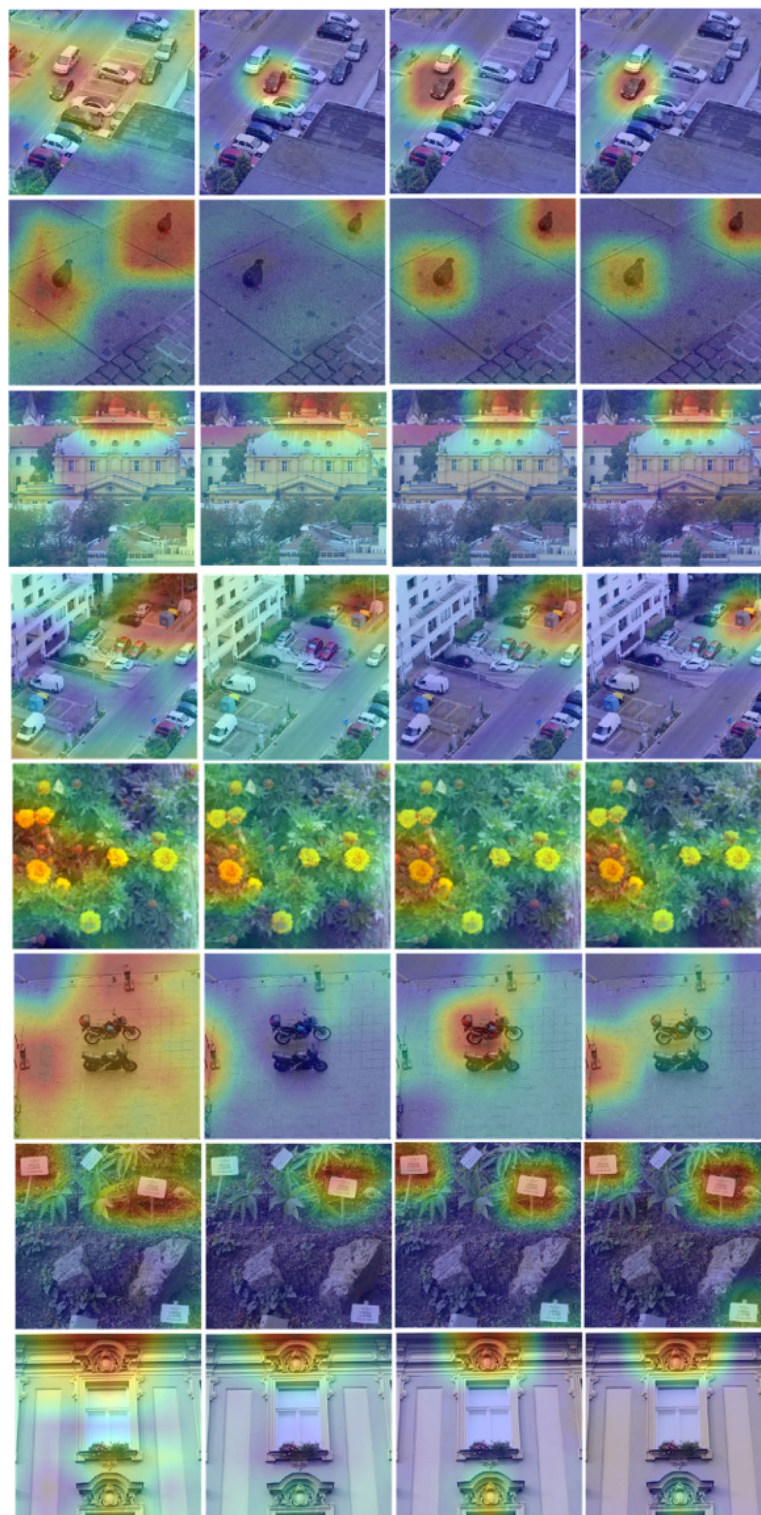


Fig. 15. Heat maps generated by MobileNet, EfficientNetB0, VGG16, and a hybrid network (from left to right).

slightly hinders the precise anchoring of hybrid descriptors. Despite these marginal drops, the model's ability to maintain an accuracy above 92% across all post-processing attacks underscores the resilience of integrating deep feature maps with genetic optimization for robust pixel-level localization.

Beyond the quantitative evaluation, the qualitative robustness of the proposed method regarding small and smooth forgery regions is demonstrated in Fig. 16 and Fig. 17. Traditional keypoint-based algorithms often struggle in textureless areas, such as a clear sky, where similar smooth segments can lead to false matches. Our approach overcomes this by utilizing the semantic heatmap branch to guide the localization process, ensuring

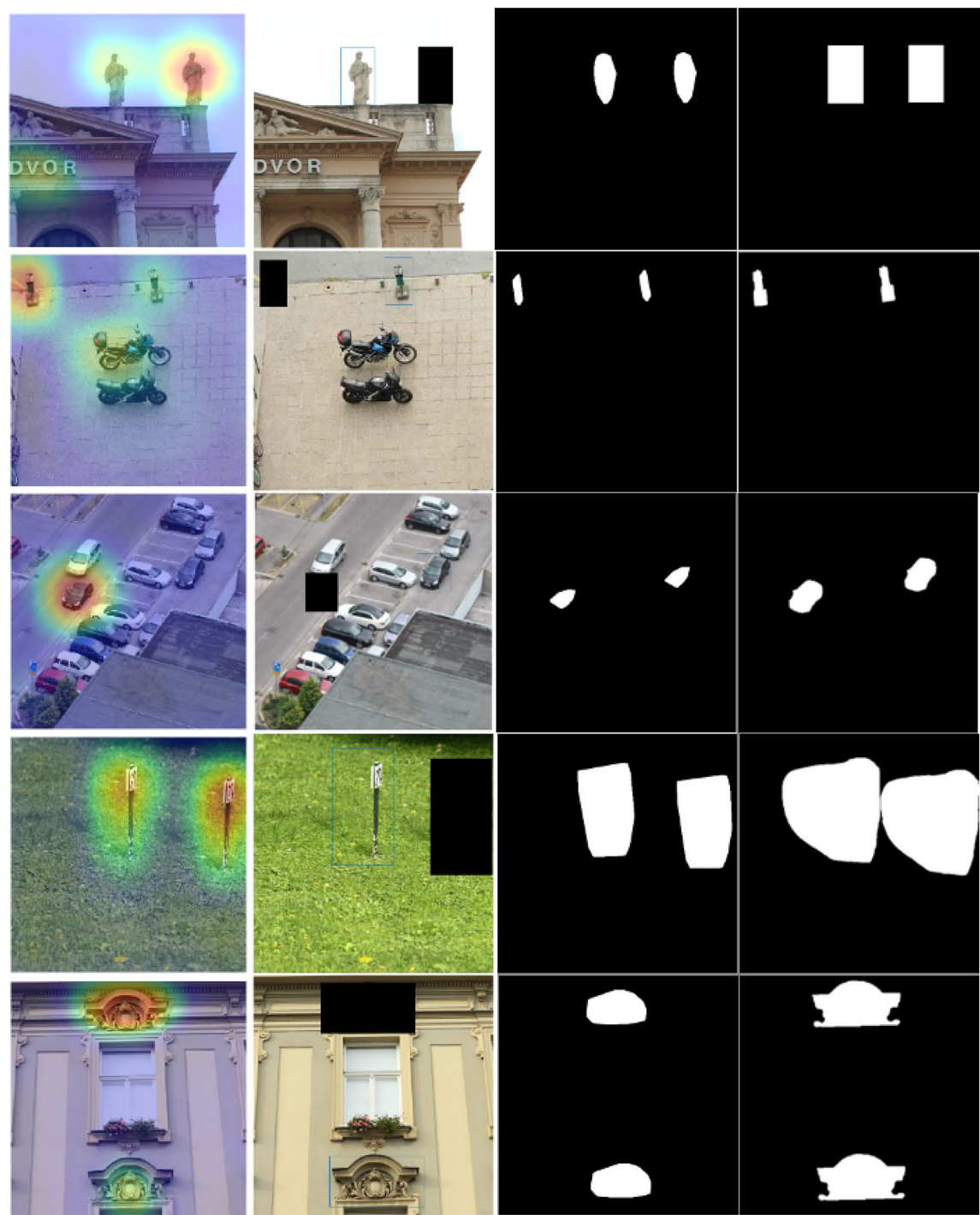


Fig. 16. Generated binary masks using hybrid model for several CoMoFod dataset images.

ACC	R	P	F1 score	FPR	IoU
97.34	91.72	92.04	91.85	2.71	87.17

Table 6. The results of the proposed model on CoMoFod dataset at pixel level.

that the detection is not solely dependent on local texture. Furthermore, as illustrated in Fig. 17, the fusion of multiple keypoint descriptors significantly increases the feature point density, which is crucial for capturing smaller forgery footprints and providing robust resistance against geometric distortions like rotation and scaling. While we acknowledge that the method’s efficiency may decrease in extremely small regions where

Post-processing	ACC	R	P	F1 score	FPR	IoU
IB1 ($\sigma^2 = 0.005$)	94.85	87.15	88.40	87.77	5.45	82.35
IB2 ($\sigma^2 = 0.0005$)	95.90	88.60	89.95	89.27	4.20	84.10

Table 7. Robustness against Blurring operations (Failure Analysis).

Post-processing	ACC	R	P	F1 score	FPR	IoU
NA1 (3×3)	93.62	85.12	86.90	86.00	6.67	80.12
NA2 (5×5)	94.75	87.05	88.10	87.57	5.32	82.20

Table 8. Robustness against Noise Addition (Failure Analysis).

Post-processing	ACC	R	P	F1 score	FPR	IoU
JC1 (QF=20)	92.15	82.35	84.80	83.55	8.92	78.45
JC2 (QF=30)	93.40	84.10	86.15	85.11	7.15	79.80
JC7 (QF=80)	95.12	87.80	89.10	88.44	4.95	83.15
JC8 (QF=90)	95.85	88.70	90.05	89.37	4.10	84.20

Table 9. Robustness against JPEG Compression (Failure Analysis).

Post-processing	ACC	R	P	F1 score	FPR	IoU
CA1 (0.01, 0.95)	95.65	88.35	89.10	88.72	4.35	83.75
CA2 (0.01, 0.90)	95.10	87.80	88.45	88.12	4.90	82.90
CA3 (0.01, 0.80)	94.10	86.15	87.50	86.82	5.85	81.40

Table 10. Robustness against Contrast Adjustments (Failure Analysis).

keypoints are insufficient, the current integration ensures a high level of precision for the majority of challenging cases in the dataset. Furthermore, while the current evaluation is performed on the CoMoFoD dataset, the integration of mathematically-derived keypoint descriptors ensures that the localization process is driven by geometric consistency rather than dataset-specific artifacts. This hybrid approach provides the model with inherent generalization capabilities, making it robust against the diverse image characteristics found in different forensic datasets.

Comparison and discussion

In most studies, block-based and keypoint-based methods are usually used to detect the forgery region, while deep learning methods and pre-trained networks are often utilized for image-level forgery detection. Therefore, a comprehensive comparison of the proposed method with other deep learning approaches is provided in Table 11. Additionally, it should be noted that the proposed method has also been evaluated on the MICC-2000 dataset, and the identified counterfeit areas are illustrated for several images from this dataset in Fig. 18. As shown in these visual results, the proposed model effectively generates precise heatmaps that pinpoint the forged regions, providing a robust localized focus even in the presence of various geometric transformations.

It is important to emphasize that the MICC-2000 dataset primarily focuses on geometric transformations (such as rotation and scaling) and does not include any post-processing or binary masks. The observed performance in Table 11 is influenced by the dataset's imbalanced nature, consisting of approximately 1,300 original and 700 forged images. Despite the lack of ground truth masks, our model successfully pinpoints the forgery regions using only the generated heatmaps, demonstrating its robustness against complex geometric distortions without relying on additional image refinements.

Also, the comparison of the proposed method for pixel forgery detection with a block-based method⁵⁰, a keypoint-based method⁵¹, two hybrid method of block-based and key point based methods^{41,52}, and two deep learning methods^{53,54} is provided in Table 12.

Based on the results in Tables 11 and 12, the block-based method is time-consuming and complex. Adjusting the block size, selecting an appropriate feature extraction algorithm, and comparing blocks to detect forgery are challenging tasks. Additionally, this method does not perform well against geometric transformations like rotation and scaling. In contrast, the keypoint-based method faces challenges in selecting a suitable keypoint extraction algorithm and determining the threshold value for a sufficient number of keypoints to detect forgery regions. This is especially difficult in smooth and small regions. Unlike these traditional methods, deep learning integrates feature extraction and forgery detection under various conditions using different architectures.

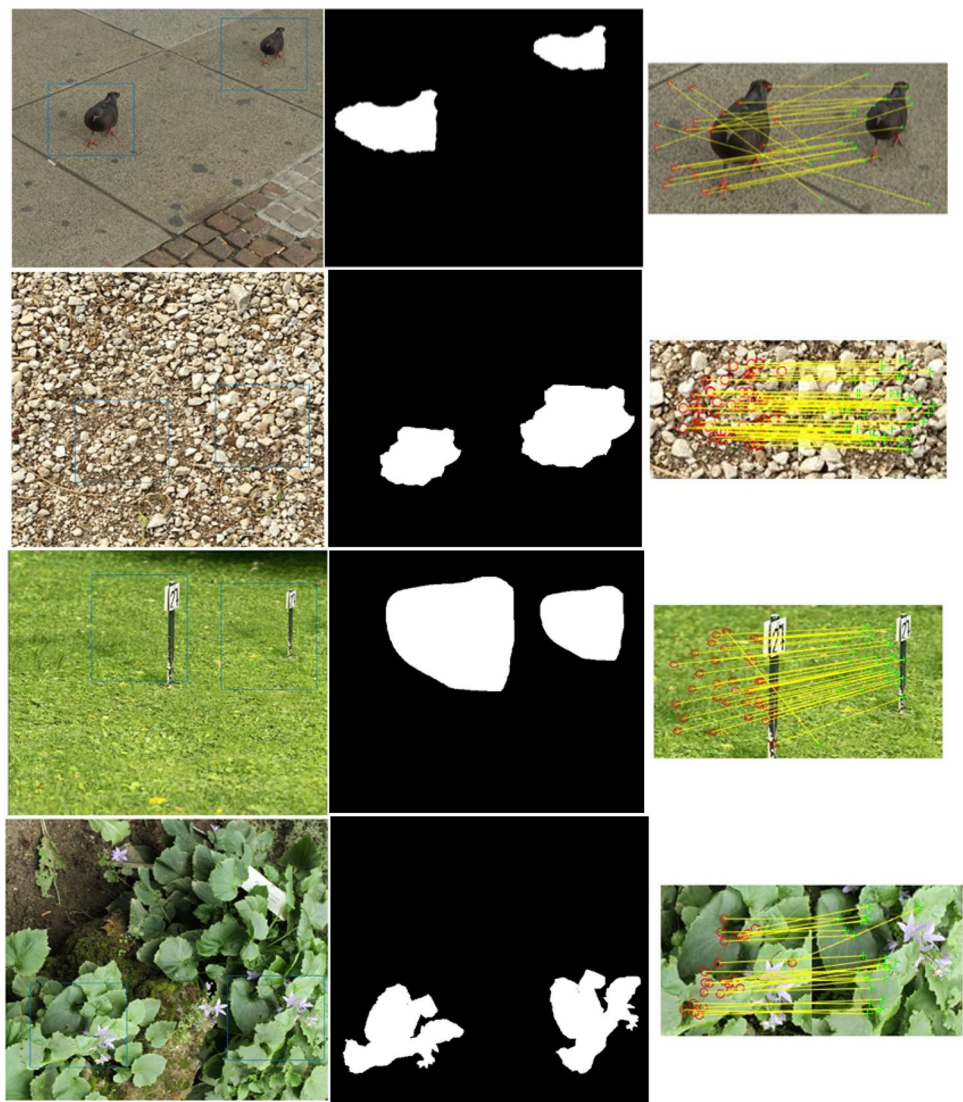


Fig. 17. Robustness of the proposed HDBK model against geometric transformations and small forgery regions utilizing dense keypoint fusion.

Method	Accuracy	Loss Function
Pre-trained AlexNet+SVM ⁴⁴	94.46	0.1689
Custom CNN network ⁴⁵	92.60	0.1956
Pre-trained MobileNetv2 ⁴⁶	93.38	0.1737
Pre-trained VGG16+GAP layer ²⁵	98.50	0.0310
Custom CNN network and KNN ⁴⁷	95.46	0.1027
ResNet network ⁴⁸	98	0.082
Graph Convolutional Networks Feature Extraction+SVM ⁴⁹	0.78	0.16
Pre-trained Mobilenet + GAP layer	95.45	0.1527
Pre-trained EfficientNetB0+GAP layer	97.40	0.0486
Vision Transformer (ViT)	98.23	0.0443
API model (MobileNet+EfficientNetb0, VGG16) on MICC2000 dataset	88.89	0.5891
API model (MobileNet+EfficientNetb0, VGG16)	98.76	0.0286

Table 11. Comparison of the proposed model with others at image level.

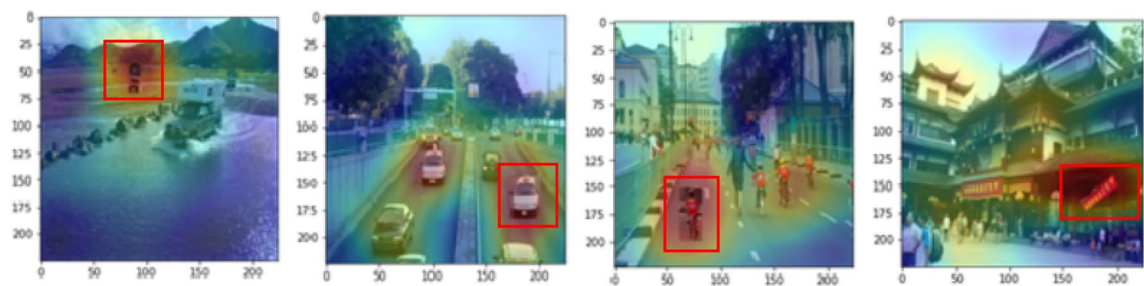


Fig. 18. Heatmap generated by proposed method on MICC 2000 dataset.

Method	ACC	P	R	F1 score	FPR	IoU
Block-based method (DCT(8 × 8blocks)) ⁵⁰	95	99.21	82.26	89.94	22.79	75.73
Key point-based method (DWT + SIFT) ⁵¹	90.11	75.71	100	86.17	48.49	86.93
A hybrid of block-based and key point-based methods(DCT+ORB) ⁵²	85.5	84.31	87	85.58	24.19	84.82
A hybrid of block-based and key point-based methods using meta heuristic algorithms ⁴¹	95.05	90.15	91.80	90.96	16.34	78.81
BusterNet ⁵³	90.86	57.34	49.39	49.26	35.67	50.77
DOA-GAN ⁵⁴	80.67	54.70	39.67	41.44	29.83	49.54
A hybrid of block-based and key point-based methods using SLIC Segmentation ⁵⁵	93.25	93.78	93.74	93.75	7.25	86.24
Proposed model	97.34	91.72	92.04	91.85	2.71	87.17

Table 12. Comparison of the proposed model with others at pixel level.

However, the biggest disadvantage of deep learning is the need for a large dataset for training, which is often lacking in the field of forgery detection.

In the proposed method, a hybrid approach is utilized to leverage the complementary advantages of deep learning, keypoint descriptors, and meta-heuristic optimization. Forgery detection at the image level is achieved through a triple-architecture ensemble for precise heatmap generation, which provides initial resilience against geometric transformations. As illustrated in Fig. 15, the integration of feature maps to generate a heatmap effectively identifies suspected forgery regions, eliminating the dependency on ground-truth datasets for training. The overall implication of this hybrid strategy is a significant reduction in the search space for the Genetic Algorithm, which serves as an optimization tool to prune the search space and avoid a blind exhaustive search. By focusing the optimization process on a limited number of candidate states within the suspected regions—rather than performing a blind exhaustive search across the entire image—the average runtime is significantly improved compared to traditional methods. Specifically, the processing time is divided into 14.35 seconds for the deep learning-based heatmap generation and 12.6 seconds for the Genetic Algorithm-optimized search. This reduces the total time from approximately 51.12 seconds in traditional block-based methods to a much more efficient duration.

This efficiency directly contributes to a superior localization accuracy (97.34%) and a remarkably low false positive rate (2.71%). Furthermore, the performance analysis highlights that the synergy between hybrid keypoint descriptors (SURF, FAST, and BRISK), which are integrated for finer discovery of forged areas and providing high robustness against geometric transformations such as rotation and scaling, and the deep feature heatmap ensures high attack resistance. Specifically, these keypoints allow the model to maintain structural consistency even under complex rotations and scaling scenarios where purely pixel-based or active forensic methods may struggle or require pre-embedded data. This balance between computational efficiency and robust detection makes the HDBK model a reliable tool for real-world passive forensic applications.

Limitations and future research directions

Despite the superior performance and efficiency of the HDBK model, certain limitations should be acknowledged. While the hybrid approach is highly effective in detecting small-scale forgeries where traditional keypoint-based methods often fail due to low feature density it faces challenges in textureless or monochromatic regions (e.g., clear skies). In such cases, the sparsity of keypoints can hinder the precise anchoring of the descriptors. Furthermore, as shown in the failure analysis, extreme post-processing such as heavy JPEG compression ($QF = 20$) remains a challenging boundary for the Genetic Algorithm’s similarity search. In future work, we plan to extend the framework to address multi-type forgeries, including image splicing and object removal, by integrating noise residual analysis. Additionally, we aim to enhance the keypoint matching stage by incorporating Transformer-based architectures or deep-learned descriptors (e.g., SuperPoint). These advancements are expected to improve the model’s reliability in smooth regions and enhance its robustness against more sophisticated geometric transformations.

Conclusion

Image forgery detection remains a significant challenge due to the complexity of identifying pixel-level manipulations in diverse scenarios. To address this, we introduced the HDBK model, a novel hybrid framework designed to tackle major challenges such as detection in smooth regions, robustness against geometric transformations, and handling multiple forgeries. By integrating image-level and pixel-level analysis, HDBK provides a robust solution for accurate copy-move forgery detection.

The proposed model operates in three integrated parts. First, the image-level detection phase innovatively constructs a heatmap of suspected regions using pretrained deep networks, where the final layers were specifically fine-tuned on the CoMoFoD dataset to optimize localization accuracy. Second, a pixel-level identification phase utilizes an improved block-based method combined with a Genetic Algorithm to identify similar blocks. Finally, a matching phase using keypoint extraction is employed to create an accurate convex polygon of the forged pixel regions, ensuring precise anchoring and source-target identification.

Experimental results on the CoMoFoD dataset demonstrate that HDBK outperforms existing methods, showing high robustness to post-processing operations like blurring, noise addition, and JPEG compression. By leveraging high-level semantic features, the model significantly improves detection in smooth areas, though identifying very small forged regions with low keypoint density remains a challenge. While the hybrid architecture requires more memory than single-stream models, the heatmap-guided strategy ensures high computational efficiency for forensic applications. Although the current investigation focuses on the robust localization of copy-move forgeries, the hybrid nature of the HDBK model provides a solid foundation for detecting other types of manipulations. For future work, we plan to adapt the genetic search and keypoint matching strategies to identify inconsistencies across multiple image sources, thereby extending the framework to detect sophisticated forgeries such as image splicing and AI-driven object removal. Additionally, investigating lightweight architectures to further enhance processing speed, adapting the framework for video forgery detection, and evaluating its robustness against adversarial attacks are key directions for our future research.

Data availability

The publicly available dataset utilized in this study (CoMoFoD dataset) can be accessed directly at: <https://www.vcl.fer.hr/comofod/comofod.html>. Code reference <https://zenodo.org/records/18414848>.

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

This study is based entirely on the analysis of the publicly available CoMoFod (Copy-Move Forgery Detection) dataset. The dataset consists of computer-generated and manipulated images, and does not contain any images of human participants, animals, or private/personal data. Therefore, this research did not involve human subjects or animal experimentation. According to our institutional guidelines and international standards, this type of research does not require ethics committee approval or informed consent from participants.

Animal ethics

Not applicable. No animal data or experimentation was involved in this study.

Consent for publication

Not applicable. No identifying information or images of any individual are included in this manuscript.

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